

Binomial, Bernoulli & Poisson Distributions

Clear study notes with properly worded questions, step-by-step solutions, interpretations, and charts

Examples use practical e-commerce and customer-service situations.

Binomial → Bernoulli → Poisson → Comparison → Formula Sheet

PART 1 — BINOMIAL DISTRIBUTION

1. What is a Binomial Distribution?

A **binomial distribution** is used to find the probability of obtaining a particular number of successes in a fixed number of independent trials. Each trial has only two possible outcomes: success or failure.

E-commerce example: An online store sends a promotional offer to 5 customers. For each customer, the probability that the customer places an order is 60%. Let X represent the number of customers who place an order.

2. Conditions for a Binomial Distribution

Condition	Meaning
Fixed number of trials (n)	The promotion is sent to a fixed number of customers.
Two outcomes	Each customer either places an order (success) or does not place an order (failure).
Independent trials	One customer's decision does not affect another customer's decision.
Constant probability (p)	Each customer has the same 60% probability of placing an order.

3. Binomial Formula

Question: What is the probability that exactly x successes occur in n independent trials?

Formula: $P(X = x) = C(n, x) p^x (1-p)^{n-x}$

where n = total trials, x = required successes, p = success probability, and 1-p = failure probability.

4. Mean, Variance and Standard Deviation

Measure	Formula	Interpretation
Mean	np	Expected number of successes in the n trials.
Variance	$np(1-p)$	Measures the variability in the number of successes.
Standard deviation	$\sqrt{np(1-p)}$	Measures the typical spread of the number of successes around the mean.

5. Numerical Question

Question: An e-commerce company sends a promotional offer to 5 customers. The probability that any one customer places an order is 0.60. Assuming the customers' decisions are independent, what is the probability that exactly 3 of the 5 customers place an order?

Solution:

n = 5, x = 3, p = 0.60, and 1-p = 0.40

$$\begin{aligned} P(X=3) &= C(5,3)(0.60)^3(0.40)^2 \\ &= 10 \times 0.216 \times 0.16 \\ &= \mathbf{0.3456 = 34.56\%} \end{aligned}$$

Quantity	Calculation	Answer
p	Given	0.60 (60%)

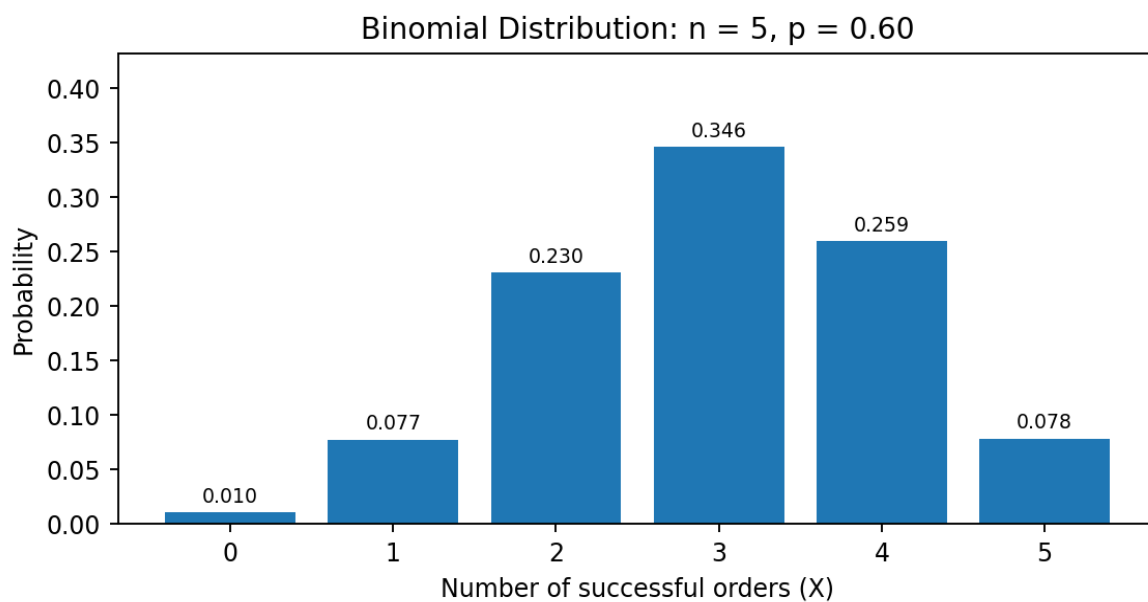
Quantity	Calculation	Answer
1-p	$1 - 0.60$	0.40 (40%)
$P(X=3)$	$C(5,3)(0.60)^3(0.40)^2$	0.3456 (34.56%)
Mean	$np = 5 \times 0.60$	3 orders
Variance	$np(1-p) = 5 \times 0.60 \times 0.40$	1.20
SD	$\sqrt{1.20}$	1.095 orders

6. Interpretation

$P(X=3) = 34.56\%$: There is a 34.56% probability that exactly 3 of the 5 customers place an order.

Mean = 3 orders: If this same type of promotion were repeated many times with groups of 5 customers, the average number of orders per group would be 3.

SD = 1.095 orders: The number of orders in groups of 5 customers typically varies around the mean of 3, with a standard deviation of about 1.095 orders.



PART 2 — BERNOULLI DISTRIBUTION

1. What is a Bernoulli Distribution?

A **Bernoulli distribution** describes a single trial that has exactly two possible outcomes. We usually code success as $X=1$ and failure as $X=0$.

E-commerce example: Consider one customer who receives a promotional offer. The customer either places an order or does not place an order. Suppose the probability of placing an order is 70%.

2. Bernoulli Probability Formula

Question: What is the probability that the single trial results in x ?

Formula: $P(X=x) = p^x(1-p)^{1-x}$, where $x = 0$ or 1 .

Outcome	X	Probability
Customer does not order	0	$P(X=0) = 1-p$
Customer places an order	1	$P(X=1) = p$

3. Mean, Variance and Standard Deviation

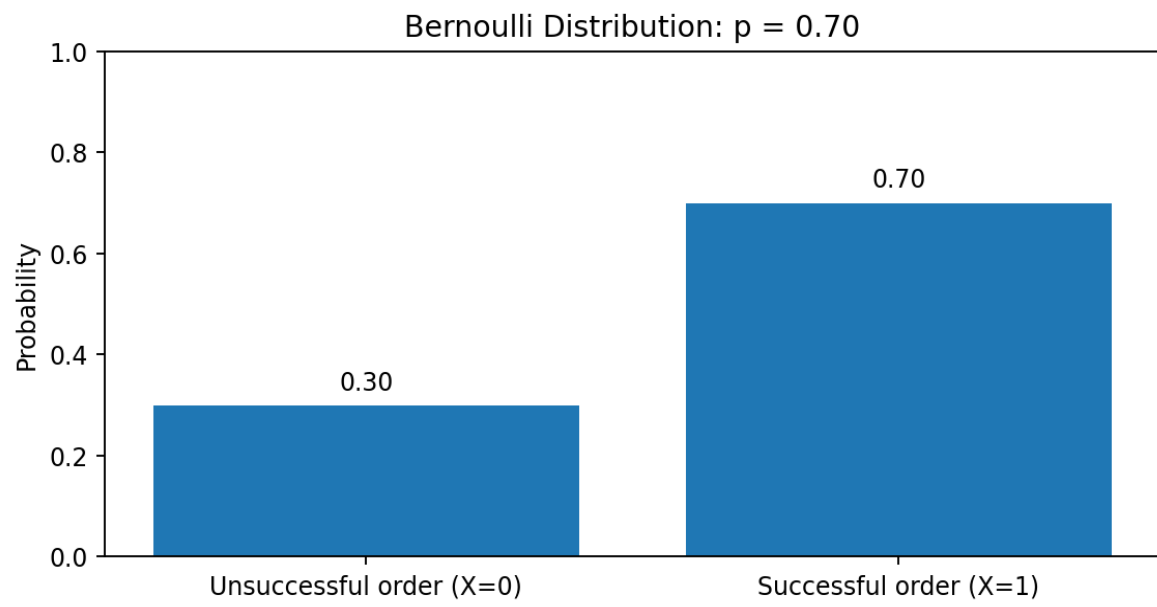
Measure	Formula	Interpretation
Mean	p	Expected value of the 0/1 outcome.
Variance	$p(1-p)$	Measures variability in the binary outcome.
Standard deviation	$\sqrt{p(1-p)}$	Measures the spread of the binary random variable around its mean.

4. Numerical Question

Question: An online store estimates that a randomly selected customer has a 70% probability of placing an order after receiving a promotional offer. Define $X=1$ if the customer places an order and $X=0$ otherwise. Find p , $1-p$, the mean, the variance, and the standard deviation.

Quantity	Calculation	Answer
p	Given	0.70 (70%)
$1-p$	$1 - 0.70$	0.30 (30%)
Mean	p	0.70
Variance	$p(1-p)$	0.21
SD	$\sqrt{0.21}$	0.458

Interpretation: Before observing the customer's decision, the model assigns a 70% probability to an order and a 30% probability to no order. The mean is 0.70 because the outcome is coded as 0 or 1. The SD of 0.458 is a mathematical measure of the uncertainty/variability of this binary random variable; it is not a 45.8% probability and it does not mean the customer's actual result is 0.458.



PART 3 — POISSON DISTRIBUTION

1. What is a Poisson Distribution?

A **Poisson distribution** is used to model the number of times an event occurs in a fixed interval of time, distance, area, or another specified interval, when events occur independently and the average rate is reasonably constant.

Customer-service example: An e-commerce website receives customer-support messages at an average rate of 3 messages per hour. Let X be the number of messages received during one hour.

2. Poisson Formula

Question: What is the probability of observing exactly x events during the specified interval?

Formula: $P(X=x) = e^{-\lambda} \lambda^x / x!$

Here, λ is the average number of events in the interval and $x = 0, 1, 2, \dots$

3. Mean, Variance and Standard Deviation

Measure	Formula	Interpretation
Mean	λ	Expected/average number of events in the interval.
Variance	λ	Measures variability in the event count.
Standard deviation	$\sqrt{\lambda}$	Measures the typical spread around the mean.

4. Numerical Question

Question: An e-commerce company's customer-support system receives an average of 3 messages per hour. Assuming a Poisson model is appropriate, what is the probability that exactly 2 messages are received in the next hour? Also find the mean, variance, and standard deviation.

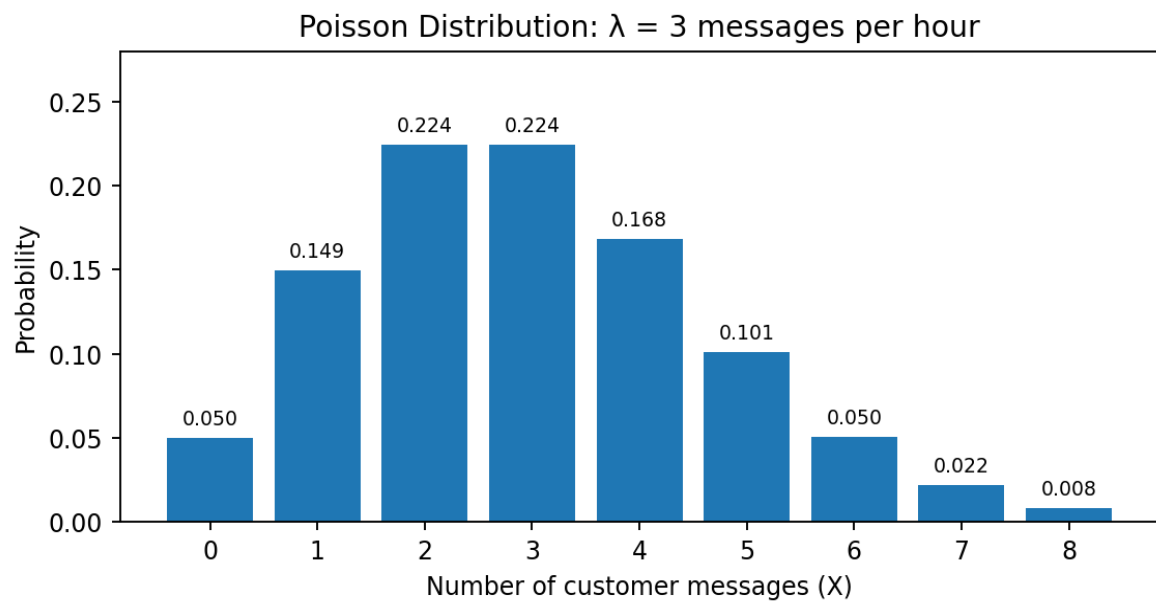
Solution:

$\lambda = 3$ and $x = 2$

$P(X=2) = e^{-3} \times 3^2 / 2!$
= 0.2240 = 22.40%

Quantity	Calculation	Answer
λ	Average messages per hour	3
$P(X=2)$	$e^{-3} \times 3^2 / 2!$	0.2240 (22.40%)
Mean	λ	3 messages/hour
Variance	λ	3
SD	$\sqrt{3}$	1.732 messages

Interpretation: There is approximately a 22.40% probability of receiving exactly 2 customer-support messages in one hour. The expected number is 3 messages per hour. The SD of 1.732 messages indicates the typical spread of the hourly message count around its mean of 3.



PART 4 — BINOMIAL vs. BERNOULLI vs. POISSON

Feature	Bernoulli	Binomial	Poisson
Main idea	One yes/no trial	Count successes in fixed n trials	Count events in a fixed interval
Possible values of X	0 or 1	0 to n	0,1,2,...
Parameters	p	n and p	λ
Mean	p	np	λ
Variance	$p(1-p)$	$np(1-p)$	λ
SD	$\sqrt{p(1-p)}$	$\sqrt{np(1-p)}$	$\sqrt{\lambda}$
Example	One customer orders or not	Orders from 5 customers	Support messages in 1 hour

Easy way to remember

Bernoulli: ONE trial → Did the customer order?

Binomial: FIXED number of trials → How many customers ordered out of 5?

Poisson: FIXED interval → How many messages arrived in one hour?

PART 5 — FINAL FORMULA SHEET

Bernoulli: $P(X=x)=p^x(1-p)^{1-x}$; Mean=p; Variance= $p(1-p)$; SD= $\sqrt{p(1-p)}$.

Binomial: $P(X=x)=C(n,x)p^x(1-p)^{n-x}$; Mean=np; Variance= $np(1-p)$; SD= $\sqrt{np(1-p)}$.

Poisson: $P(X=x)=e^{-\lambda}\lambda^x/x!$; Mean= λ ; Variance= λ ; SD= $\sqrt{\lambda}$.